

The Lazaros–Eudora Method (LEM): A Topological Framework for User–LLM Interaction Dynamics

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Abstract

We introduce the Lazaros–Eudora Method (LEM), a conceptual and computational framework for analyzing user–LLM interactions as dynamic trajectories in latent representation space rather than as isolated prompt–response pairs. The central hypothesis of LEM is that repeated interaction between a user and a language model induces structured trajectory regimes that can be described through user-specific attractors and compact interaction signatures. We propose both geometric and topological summaries as candidate measurements; whether topology adds value beyond appropriately matched geometric controls is an open question. We present two synthetic pipeline checks. The first demonstrates geometric identifiability of user-conditioned trajectories in a controlled latent dynamical system, achieving perfect nearest-neighbor re-identification in the reported synthetic setting across $N = 500$ users and ten independent random seeds, compared to a random baseline of 0.002. The second shows that convergent and cyclic interaction regimes can be distinguished through persistent H_1 structure in one constructed trajectory pair after noisy projection, yielding a descriptive ratio of 161.6 at seed 1337. Neither example establishes user signatures in language-model activations, population-level performance, or a unique advantage for topology. LEM remains a research program whose empirical claims require independent real-model tests.

1 Introduction

Current large language model evaluation remains dominated by static, turn-local metrics. These metrics can assess answer quality, benchmark performance, or stylistic alignment, but they do not directly describe the temporal structure of repeated user–model interaction. As a result, an important object remains under-modeled: the trajectory induced by a specific user interacting with a model across multiple turns.

The Lazaros–Eudora Method (LEM) is a framework designed to study this object. Its central premise is that user–LLM interaction should be modeled as a dynamical trajectory in latent space rather than as a collection of isolated prompt embeddings or output tokens. In this view, repeated interaction may generate structured regimes—for example convergence, recurrence, or instability—that are more naturally described in the language of dynamical systems and topology than in the language of static vector similarity alone.

This paper makes a limited but concrete contribution. It does not claim to establish a full theory of human cognition or a validated description of real frontier-model internals. Instead, it contributes:

1. a compact conceptual formalization of LEM;
2. a distinction between geometric and topological notions of interaction signatures;
3. two toy-model pipeline demonstrations showing how constructed user vectors and trajectory classes can be measured in controlled synthetic settings.

The intended contribution is therefore methodological: to provide a testable framework for studying user–LLM interaction dynamics and to identify which parts of the theory are already operational, which remain open, and which are presently speculative.

2 Conceptual Framework

2.1 Interaction trajectories

We define an interaction trajectory as a time-ordered sequence of states

$$x_1, x_2, \dots, x_T \in \mathcal{X},$$

where $\mathcal{X}$ is a latent or constructed representation space associated with a multi-turn user–LLM interaction. In practice, $\mathcal{X}$ may refer to internal activations, reduced coordinates, or synthetic latent states in a toy model. For inference with frozen weights, the activation at a turn depends on the supplied context. Returning to the same weights with empty history does not retain an earlier dialogue without explicit memory or training. Context-carried signatures must be distinguished from changes stored in model parameters.

This shift from isolated states to trajectories is the first core move of LEM. The object of study is not merely the embedding of a prompt or response, but the path induced by repeated interaction.

2.2 Cognitive signatures

A *cognitive signature* is a compact representation derived from an interaction trajectory that preserves salient structural properties of user–model dynamics. In the present paper, we distinguish two operational levels:

1. **Geometric signatures**, defined through distances, directional similarity, and nearest-neighbor re-identification in a controlled latent system;
2. **Topological signatures**, defined through persistent homology, especially the presence or absence of robust H_1 features.

This distinction is a proposed measurement strategy. A topological summary captures loops in a point cloud, but the present examples do not show that it outperforms geometric summaries of the same trajectories.

2.3 Attractor-like interaction regimes

LEM uses the term *attractor* in a modeling sense. The claim is not that real human cognition has been recovered from LLM traces, but that repeated interaction can be approximated as a dynamical process whose trajectories may approach stable regimes. In the minimal setting considered here, these include:

- **Convergent regimes**, approximated by a stable fixed point;
- **Cyclic regimes**, approximated by a stable limit cycle;
- **Unstable or noisy regimes**, in which no compact structural summary is stable.

2.4 Core hypotheses

We state four hypotheses. The synthetic examples instantiate aspects of the first three by construction; none is empirically established for real user-LLM interaction by these examples.

H1: Geometric identifiability. User-conditioned interaction dynamics can generate trajectories that are re-identifiable above chance in a controlled latent system.

H2: Regime differentiation. Convergent and cyclic interaction regimes generate structurally distinct trajectory classes.

H3: Topological utility. Persistent homology captures structural differences between such regimes, especially through the presence or absence of persistent H_1 features.

H4: Cross-version stability. User-specific interaction signatures may remain partially stable across related model variants or model families. In this paper, H4 is treated as a theoretical target rather than an empirically established claim.

3 From Geometric Similarity to Topological Structure

Many current approaches to personalization, user modeling, and dialogue adaptation rely on static similarity measures, clustering, or compressed user embeddings. These approaches are useful, but they typically summarize multi-turn interaction as a point estimate or a low-order statistic.

LEM instead treats interaction as a trajectory and therefore distinguishes between two levels of structure:

1. **Geometric structure:** whether trajectories lie near each other or near user-conditioned attractors;
2. **Topological structure:** whether trajectories exhibit recurrence, loop-like organization, or topological triviality.

This motivates the use of topological data analysis (TDA), especially persistent homology. In the current work, the key topological observable is persistent H_1 structure, which can detect a sampled loop in a constructed point cloud. A loop alone does not establish a dynamical attractor: recurrence and return after perturbation require temporal and interventional evidence.

We explicitly do *not* claim that topological features directly reveal psychological essence or consciousness. A claim of added value over geometry requires a matched nonloop control and an independent comparison with geometric summary statistics; neither is included in the original Toy V2 example.

4 Toy-Model Pipeline Checks

The present paper does not validate LEM on real model internals. These deliberately constructed examples check that the numerical pipeline behaves as expected under known dynamics. An implementation defect could make such a check fail. Success, however, does not provide a test of whether user-specific signatures or attractors exist in language models. That question requires independent data and a falsifiable real-model design.

We therefore present two toy checks with distinct roles. The first constructs well-separated user-indexed synthetic vectors and checks their geometric identifiability. The second constructs one sink and one cycle and checks whether their H_1 lifetimes differ in this example. Neither check compares a topology method with a matched geometric baseline on held-out instances.

4.1 Toy Experiment 1: Geometric Identifiability

The first toy experiment is a geometric sanity check for the notion of user-specific interaction structure. We consider a latent dynamical system of the form

$$x_{t+1} = (1 - \alpha - \beta) x_t + \alpha s_i + \beta d_m + \varepsilon_t, \quad (1)$$

where $x_t \in \mathbb{R}^d$ denotes the latent state at time t , s_i is a user-specific signature vector, d_m is a domain or model attractor, and $\varepsilon_t \sim \mathcal{N}(0, \sigma^2 I_d)$ is isotropic Gaussian noise. The interpretation is deliberately minimal: s_i models a user-conditioned directional bias in latent space, while d_m models task- or model-level regularization toward a shared region. This is a controlled synthetic separability regime and does not simulate a deployed user-model interaction.

4.1.1 Setup

We instantiate $N = 500$ users and $M = 2$ model domains in a $d = 512$ -dimensional embedding space. User vectors and initial domain vectors are independently drawn from $\mathcal{N}(0, I_d)$ and ℓ_2 -normalised. The second domain is then replaced by $d_1 = \text{normalize}(d_0 + 0.4z)$, with a fresh $z \sim \mathcal{N}(0, I_d)$; the two final domains are therefore not independent constructions. Trajectories use $\alpha = 0.4$, $\beta = 0.3$, $\sigma = 0.15$, and $T = 200$. Empirical signatures retain the published definition:

$$\hat{s}_i^{(m)} = \frac{1}{K} \sum_{j=0}^{K-1} x_{\tau-j}^{(i,m)} - \beta d_m, \quad K = 40. \quad (2)$$

Here τ denotes the last stored state index, so exactly K states are averaged. Vectorized training performs T updates ($\tau = T$); scalar test trajectories store the initial state and $T - 1$ updates ($\tau = T - 1$). These original execution conventions are preserved. The deterministic fixed point of Eq. (1) is

$$x^* = \frac{\alpha s_i + \beta d_m}{\alpha + \beta}. \quad (3)$$

In the noise-free limit, the retained signature estimator is thus $(4/7)s_i + (9/70)d_m$. Subtracting βd_m does not fully remove the fixed point’s domain component. Cross-domain cosine values describe

this operational signature, including residual domain dependence, rather than a purely user-specific estimate.

The scaled experiment uses ten seeds: 42, 123, 999, 2024, 7, 314, 1337, 77, 256, 88. The robustness experiment below instead uses one fixed seed.

4.1.2 Metrics

We evaluate three observables: (i) mean ℓ_2 distance between the empirical mean of the last K states and the theoretical fixed point of Eq. (3); (ii) nearest-neighbor re-identification accuracy over $N_{\text{test}} = 500$ fresh test queries per model; (iii) same-user versus different-user cosine similarity of empirical signatures across model domains.

4.1.3 Results

Table 1 reports summary statistics across all ten seeds. Standard deviations are sample standard deviations (ddof = 1), calculated from the six-decimal per-seed exports, as in the original Pandas aggregation. The corrected distance uses Eq. (3); the archival implementation mistakenly omitted its denominator. NN accuracies and cosine values are unchanged in a matched before/after run. The source and numerical provenance are recorded in `CORRECTION_NOTE.md` and the accompanying result manifest.

Table 1: Toy Experiment 1 — scaled results in the controlled synthetic separability regime ($d=512$, $N=500$, $T=200$, 10 seeds). Std denotes sample SD (ddof = 1).

Metric	Mean	Std	Min	Max
Random baseline accuracy	0.0020	—	—	—
NN re-id accuracy	1.0000	0.0000	1.0000	1.0000
Mean attractor distance	0.7597	0.0008	0.7583	0.7607
Cosine (same user)	0.3570	0.0020	0.3538	0.3609
Cosine (different user)	0.0022	0.0018	-0.0011	0.0056

Nearest-neighbor re-identification accuracy reaches 1.000 ± 0.000 across all seeds—a $500\times$ improvement over the random baseline of 0.002 in this synthetic setting. Same-user cosine similarity (0.3570 ± 0.0020) is far above the near-zero cross-user similarity (0.0022 ± 0.0018), confirming geometric separability.

4.1.4 Robustness analysis

We evaluate the original grid

$$\sigma \in \{0.05, 0.15, 0.30, 0.50, 0.75, 1.00, 1.50, 2.00\}.$$

Dimensions, user/domain counts, weights, trajectory lengths and window lengths match the scaled experiment. V1b uses 300 fresh test queries per model (600 total), compared with 500 per model in the scaled experiment. Seed 42 generates the fixed user/domain setup and is reset to 42 before every noise level, preserving the original shared random sequence across levels. This is a single-seed comparison; no across-seed standard deviation is reported for V1b.

Figure 2 shows perfect re-identification at the sampled levels $\sigma = 0.05$ and 0.15 , followed by accuracy 0.2300 at 0.30 . The grid has no observation at 0.25 and does not locate a precise transition.

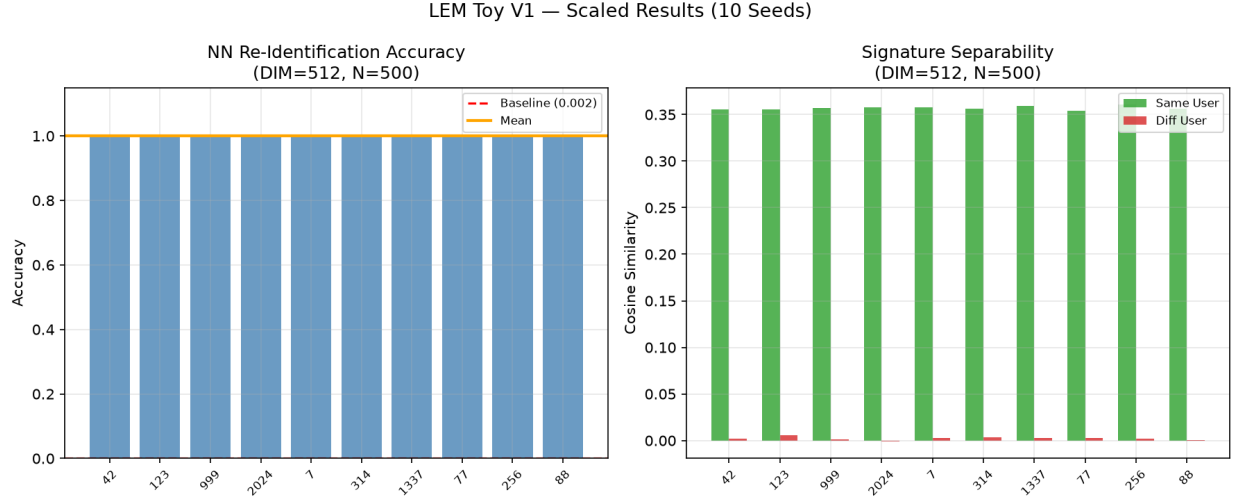


Figure 1: Toy Experiment 1. **Left:** NN re-identification accuracy across ten seeds (orange line = mean, red dashed = random baseline). **Right:** cosine similarity for same-user (green) versus different-user (red) signature pairs. All results are within the controlled synthetic separability regime described in Section 4.1.

Table 2: Robustness to noise level σ ($d=512$, $N=500$, seed = 42; 300 queries per model). Attr. Dist. is the corrected tail-mean distance.

σ	NN Acc.	Attr. Dist.	Cos. (same)	Cos. (diff)
0.05	1.0000	0.2533	0.8037	0.0045
0.15	1.0000	0.7599	0.3545	0.0019
0.30	0.2300	1.5198	0.1219	0.0006
0.50	0.0200	2.5329	0.0469	0.0002
0.75	0.0050	3.7994	0.0207	0.0001
1.00	0.0033	5.0659	0.0112	0.0000
1.50	0.0033	7.5988	0.0043	0.0000
2.00	0.0017	10.1317	0.0018	0.0000

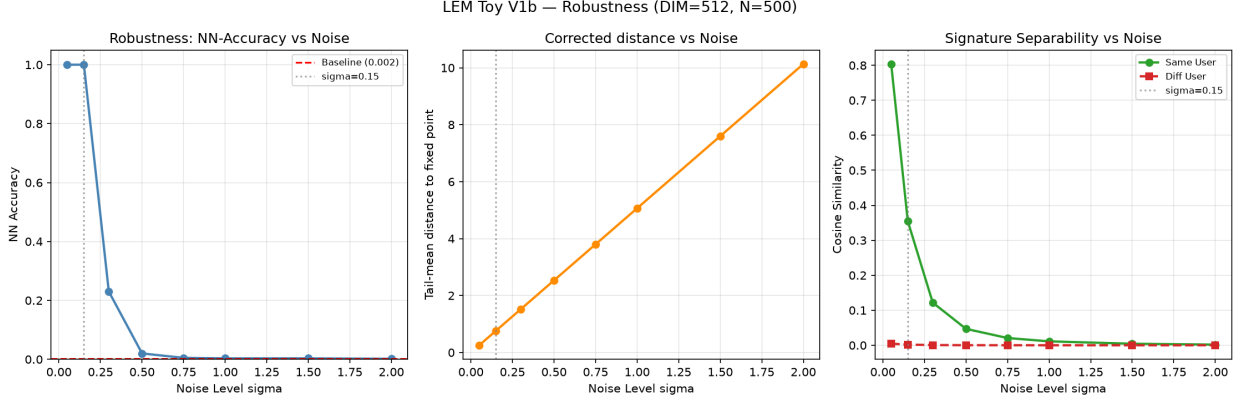


Figure 2: Robustness of LEM signatures on the sampled noise grid. **Left:** NN accuracy drops between $\sigma = 0.15$ and 0.30 ; no threshold at 0.25 was measured. **Centre:** corrected tail-mean distance to the fixed point. **Right:** decreasing same-user cosine similarity and near-zero different-user similarity.

At 0.50 , accuracy is 0.0200 , ten times the random baseline 0.0020 . At the larger sampled levels 0.75 – 2.00 , observed accuracies range from 0.0050 to 0.0017 ; the single-seed query counts do not establish exact equality to chance. Same-user cosine similarity decreases with noise while different-user cosine remains near zero. These observations alone do not isolate a causal mechanism for classification errors.

The corrected tail-mean distance is approximately proportional to σ over the sampled grid. Its magnitude must be distinguished from fluctuations of an individual state. For $\rho = 1 - \alpha - \beta = 0.3$, the stationary variance of one state coordinate is $\sigma^2/(1 - \rho^2)$. Averaging K consecutive, temporally correlated states gives per-coordinate variance

$$v_K = \frac{\sigma^2}{(1 - \rho^2)K^2} \left[K + 2 \sum_{h=1}^{K-1} (K - h)\rho^h \right].$$

Thus the stationary root-mean-square norm of the tail-mean error is $\sqrt{dv_K}$, approximately 5.069σ for $d = 512$, $K = 40$. The reported metric averages norms and is not exactly this RMS quantity; finite sampling also matters. The dependence on dimension, window length and temporal correlation precludes interpreting the observed slope as a universal single-state noise law.

Perfect re-identification here is a property of the high-dimensional synthetic construction, in which random user unit vectors are nearly orthogonal. It is not evidence of real-world user identifiability. The sampled loss of accuracy characterizes this particular simulation and does not establish a general operating threshold for LEM.

4.2 Toy Experiment 2: Topological Class Separation

The second toy experiment is an illustrative check of the persistent-homology pipeline on two prescribed dynamical systems. Its setup supplies the sink and cycle in advance; it does not test whether user interactions produce either.

4.2.1 Setup

We construct two synthetic trajectory classes:

1. a **convergent regime**, modeled as a stable sink in two dimensions:

$$\dot{x} = -\lambda x, \quad \dot{y} = -\lambda y, \quad \lambda = 0.5, \quad (4)$$

initialised at $(x_0, y_0) = (2.0, 1.5)$;

2. a **cyclic regime**, modeled as a stable limit cycle via the Van der Pol oscillator:

$$\dot{x} = y, \quad \dot{y} = \mu(1 - x^2)y - x, \quad \mu = 1.5, \quad (5)$$

initialised at $(x_0, y_0) = (0.1, 0.1)$.

Both systems are integrated over $T=40$ time units with $\Delta t=0.1$ (401 steps), then embedded into $\mathbb{R}^{16}$ via random linear projections $A \in \mathbb{R}^{2 \times 16}$ with additive Gaussian noise ($\sigma=0.05$). The executed code samples a separate projection for each trajectory; their geometry and scale are therefore not controlled to match. Persistent homology in degree H_1 is computed via the Vietoris–Rips filtration using `ripser` [3, 14].

4.2.2 Metric

For each trajectory we compute the degree-1 persistence diagram $\text{PD}(H_1)$. The key quantity is the *maximum persistence*

$$p_{\max} = \max_{(b,d) \in \text{PD}(H_1)} (d - b),$$

where b and d denote the birth and death of a homological feature. A large $p_{\max}$ summarizes the longest detected one-dimensional feature in each sampled point cloud. The historical experiment calls the following quotient its “SNR”; it is a descriptive ratio of two maxima, not an independently calibrated signal-to-noise statistic:

$$\text{SNR} = \frac{p_{\max}^{\text{cyclic}}}{p_{\max}^{\text{convergent}} + \epsilon}, \quad \epsilon = 10^{-9}.$$

4.2.3 Results

Table 3: Toy Experiment 2 — one constructed sink/cycle pair ($d=16$, $T=40$, seed = 1337); descriptive results only.

Metric	Convergent	Cyclic
Max H_1 persistence	0.0326	5.2649
Mean H_1 persistence	0.0092	0.1972
Ratio (historically “SNR”)	161.6	

The cyclic regime produces a maximum H_1 persistence of 5.265, compared to 0.033 for the convergent regime—a ratio of $161.6\times$ for this pair. A threshold $p_{\max} > 0.1$ labels these two observed examples correctly; it was not selected and evaluated on separate samples, so no error rate is estimated. There is one seed, no matched-scale noncyclic control, no held-out pairs and no comparison against simpler geometric descriptors. The large ratio therefore neither establishes that persistent homology reliably separates unseen regimes nor demonstrates that topology adds value for LEM’s user-level hypotheses.

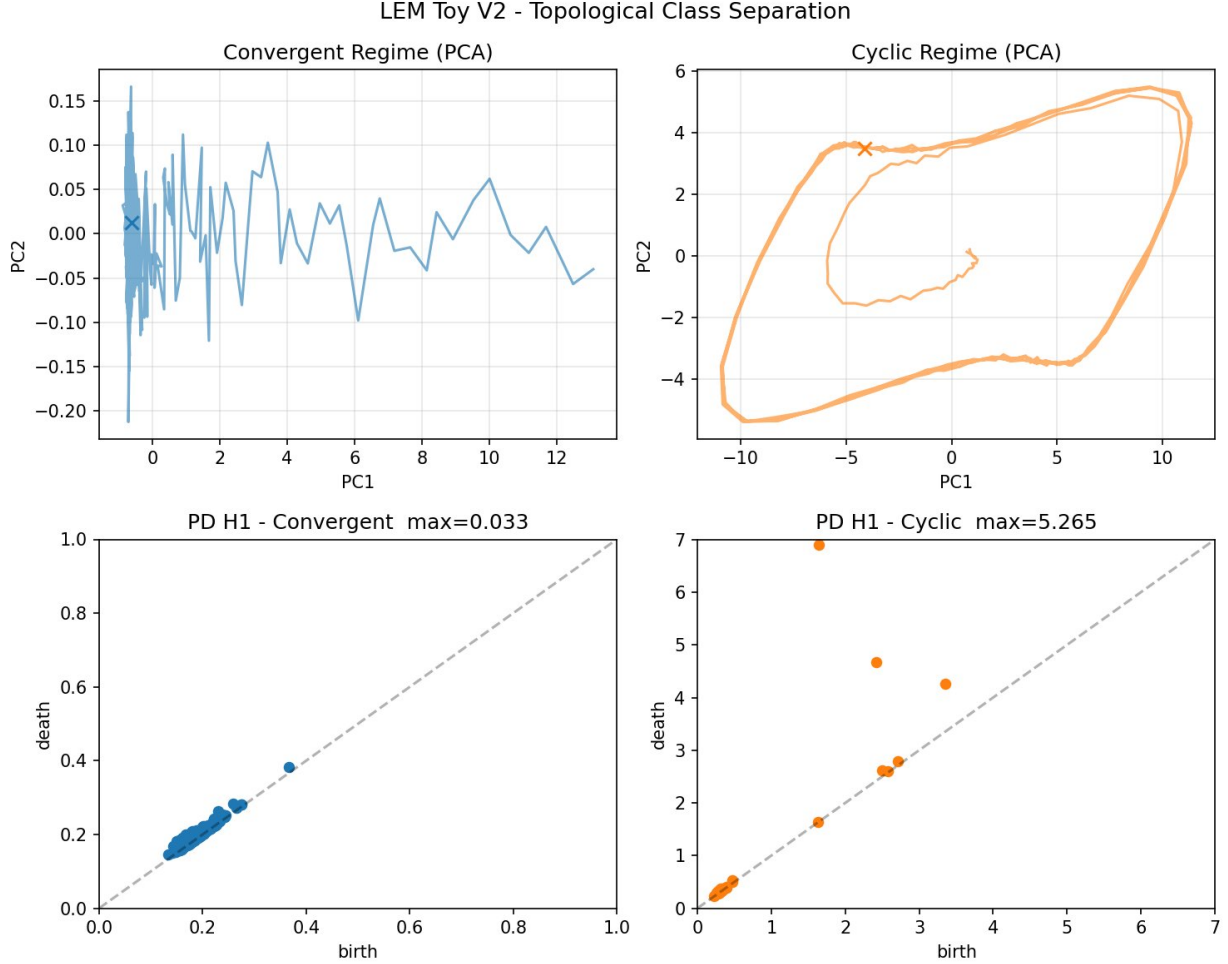


Figure 3: Toy Experiment 2: topological class separation. **Top:** PCA projections of the convergent (left, blue) and cyclic (right, orange) trajectories embedded in $\mathbb{R}^{16}$. **Bottom:** degree-1 persistence diagrams. The convergent diagram shows only diagonal noise ($p_{\max}=0.033$); the cyclic diagram contains a prominent off-diagonal generator ($p_{\max}=5.265$) for the constructed limit cycle. The historical ratio called “SNR” is 161.6 for this single pair.

4.3 What the toy experiments do and do not establish

Taken together, the two toy experiments support a restricted interpretation of LEM.

Supported in the present paper.

- User-conditioned latent trajectories can be geometrically distinguishable above chance in a controlled synthetic separability regime.
- On the one prescribed sink/cycle pair, H_1 maximum lifetimes differ; their descriptive ratio is 161.6.
- The synthetic constructions provide concrete code paths and motivate preregisterable measurements for future empirical tests.

Not established in the present paper.

- That real frontier-model activation traces exhibit the same trajectory classes.
- That interaction signatures correspond to psychological or neuroscientific identity.
- That high-dimensional synthetic separability implies real-world user identifiability.
- That cross-version stability has already been demonstrated empirically.
- That Toy V2 generalizes to new seeds or separates matched-scale controls, or that persistent homology outperforms geometric baselines.
- That LEM currently enables deployment-grade personalization, safety, or forensic attribution.

5 Related Work

LEM is related to several existing lines of research, but does not coincide exactly with any single one of them. Its closest neighbors lie in user modeling, latent behavioral representations, mechanistic interpretability, and topological analysis of neural representations.

5.1 User modeling and latent behavioral structure

A first neighboring tradition concerns latent user representations and personalization. In these approaches, user-specific state is typically modeled through compressed embeddings, memory summaries, or preference vectors. LEM is compatible with this general objective, but differs in one central respect: it treats repeated user-model interaction not primarily as a static user representation problem, but as a trajectory-level object. Related work on latent state persistence in LLMs has raised important questions about whether models maintain stable internal representations across turns at all [9]. LEM does not presuppose that real LLMs do so; rather, it asks whether the structure of user-conditioned trajectories is detectable if such persistence were present.

A closer point of contact is recent work on persona vectors. Chen et al. identify directions in activation space that correspond to behavioral or character traits—such as sycophancy, hallucination propensity, or harmfulness—and demonstrate that these directions are causally effective: injecting or suppressing them during inference predictably alters model behavior [6]. This provides direct empirical evidence that the latent spaces through which LEM posits user-conditioned trajectories

contain structured, behaviorally meaningful geometry. LEM builds on this foundation but shifts the focus from static trait directions to the temporal dynamics of repeated interaction trajectories and the possibility of deriving geometric and topological signatures from them.

5.2 Mechanistic interpretability and feature geometry

LEM is also related to mechanistic interpretability work that studies the internal organization of model representations. In particular, dictionary-learning and sparse-autoencoder work argues that model activations can be decomposed into more interpretable features than individual neurons alone [4, 13]. These results matter for LEM because they provide evidence that model representations are structured enough to support nontrivial latent analyses. A related direction is the literature on attribution graphs and circuit tracing, which models internal feature interactions as structured computational pathways [1]. LEM differs in being trajectory-centric rather than feature- or circuit-centric.

5.3 Topological analysis of representation structure

A third neighboring tradition concerns topological data analysis in machine learning and language models. The foundational framework for applying persistent homology to data analysis was established by Carlsson [5]. More recently, Gardinazzi et al. apply zigzag persistence across successive layers of LLM activation spaces [8]. Their work does not establish a general advantage over linear probes for recognizing users. Tan et al. extend this direction to reasoning traces [12]. In a complementary line of work, Chia et al. probe latent subspaces of LLMs to identify adversarial states and demonstrate that structured regions in activation space can be isolated and manipulated [7]. This supports the broader premise that LLM latent spaces contain exploitable geometric and topological structure. LEM draws directly from this methodological tradition, especially in its use of persistent H_1 structure to distinguish topologically trivial from recurrent interaction regimes.

The narrower novelty claim of LEM is therefore not that topology can be applied to neural representations in general, but that user–LLM interaction can be modeled as a dynamical trajectory whose regime class may be summarized through geometric and topological signatures.

5.4 Dynamical systems and attractor perspectives

A fourth neighboring tradition treats language models through the lens of dynamical systems. Ramsauer et al. show that Transformer attention corresponds to the update rule of modern Hopfield networks, whose energy landscape admits multiple attractor types [11]. Bai et al. demonstrate that hidden layers of deep sequence models can converge to fixed points, formalizing the notion of equilibrium computation in neural networks [2]. Most directly relevant to LEM, Wang et al. provide empirical evidence that iterative text generation in LLMs converges to attractor cycles, including two-period limit cycles under successive paraphrasing [15]. These are observations for successive paraphrasing. Ko and Geiping report model-specific attractor-like behavior in multi-turn model–model conversations [10]; they do not establish a user-specific attractor in a human–model dialogue.

LEM differs from this line of work in three respects. First, it studies user-conditioned trajectories rather than generic model dynamics. Second, it operates in latent representation space rather than in text space. Third, it employs topological analysis as a candidate way to summarize regime types. These are research aims, not experimentally established distinctions of this paper.

5.5 Positioning of the present contribution

The present paper should not be read as claiming to be the first connection between topology and language models, nor the first work on latent behavioral directions, nor the first dynamical-systems perspective on LLMs. Each of these components has established precedent. The narrower contribution of LEM is the synthesis: it introduces a single trajectory-based framework in which:

1. user-conditioned interaction is modeled dynamically (building on [2, 11, 15]);
2. attractor-like behavior becomes a first-class object of analysis, supported by evidence that latent spaces encode structured behavioral geometry [4, 6];
3. geometric identifiability and topological regime structure are treated jointly, using tools from persistent homology [5, 8].

LEM is therefore best understood as a proposed synthesis with a specific emphasis: the study of user-LLM interaction through trajectory dynamics, candidate attractor structure, and topological signatures. This review does not prove priority or uniqueness.

6 Discussion and Limitations

The main contribution of the present formulation is methodological. LEM is strongest when interpreted not as a completed explanatory theory, but as a compact research program that introduces a trajectory-based object of analysis and proposes geometric and topological tools for studying it.

The toy constructions provide two limited pipeline checks:

1. user-conditioned trajectories can be geometrically identifiable above chance in a controlled synthetic separability regime;
2. the one prescribed sink and one prescribed cycle produce different maximum H_1 lifetimes in the recorded example.

These observations do not establish that real language-model interaction traces behave in the same way. A future attractor test must specify the perturbation, recovery horizon, alternative null dynamics, and independent experimental unit before collecting confirmation data.

It bears emphasis that the perfect re-identification observed in Toy Experiment 1 is a consequence of the high-dimensional synthetic construction, not evidence for real-world user fingerprinting. In $d = 512$ dimensions, independently sampled random unit vectors are near-orthogonal by concentration-of-measure arguments, making separation trivially achievable in the absence of real-world confounds such as overlapping behavioral styles, model-induced compression, or adversarial noise. The value of Toy Experiment 1 lies not in the magnitude of the accuracy, but in confirming that the attractor-based update rule produces stable, structured trajectories in the controlled regime.

This work has several further limitations. First, all numerical illustrations use synthetic toy dynamics rather than internal activation traces from real language models; Toy V2 has one seed and two independently sampled projections and was not compared with matched geometric controls. Second, the notion of an interaction signature is operational and representational, not psychological or neuroscientific. Third, the current paper does not establish cross-version or cross-model stability. Fourth, no claim is made here that topological signatures are already useful for deployed personalization, safety, or forensic attribution. The later LEM-II-Light study on synthetic dialogues and

real model activations did not establish a prespecified incremental benefit for its activation, time, or topology blocks. Its regime-classification design did not test user-specific attractor dynamics (see [LEM-II-Light research note](#)).

The paper deliberately avoids stronger claims about cognition, consciousness, or human identity. Its scope is restricted to a formal and computational framework for analyzing interaction dynamics.

7 Future Work

There are four immediate next steps:

1. collect independent quality-checked multi-turn traces from an open-weight model, with repeated profiles, held-out topics, and explicit text controls;
2. compare independent toy sink/cycle seeds with matched-scale noncycle controls and held-out threshold selection; then compare geometric and topological summaries, including diagram distances;
3. operationalize cross-version stability through diagram-based distances or related invariants;
4. connect LEM more explicitly to current interpretability workflows, including feature-based and persona-based latent analyses.

More ambitious directions, including cross-model user continuity, privacy implications, and robust trajectory-level defenses, should be treated as future hypotheses rather than present conclusions.

The framework documentation and code are available at <https://github.com/LAKITALKS/LEM>. The archival edition v2.1.0, originally labeled “bibliography-corrected” despite errors in the archived PDF (see [ARCHIVE_ERRATUM.md](#)), is citable at [doi:10.5281/zenodo.21268033](https://doi.org/10.5281/zenodo.21268033). That DOI identifies the archive, not this unpublished correction draft.

8 Conclusion

LEM provides a compact and testable framework for analyzing user–LLM interaction dynamics as trajectories rather than isolated states. Its central contribution is to combine attractor-based intuition with a topological reformulation that makes convergent and cyclic regimes measurable in principle. The current paper presents a first version of this program: a conceptual formulation and synthetic pipeline checks. Whether these ideas hold for user-specific dynamics in model internals remains an open empirical question. However, even at the present stage, LEM provides a clearer basis for asking that question in a controlled and scientifically tractable way.

Research and tool disclosure

“Eudora” in the project name denotes AI collaboration, not a second human author. AI tools assisted with drafting, code, and review; the named author is responsible for the claims and released materials. Neither of the toy checks in this manuscript includes human participants.

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